Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Artificial Intelligence: The Gap Between Expectations and Real-World Performance

Submitted By

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Introduction

- AI investment exceeded **$180 billion** globally between 2024–2025.
- AI showed strong performance in reasoning, coding, and multimodal benchmarks.
- Organisations expected automation, faster workflows, and higher productivity.
- By 2026, many companies struggled with AI integration and measurable ROI.
- This mismatch between expectations and real-world performance is called the **AI Reality Gap.**
- Human supervision remains necessary due to AI errors and hallucinations.
- Poor-quality and unstructured enterprise data reduces AI reliability.
- Organisational resistance and workflow adaptation slow successful AI adoption.

Methodology

The study uses two case studies to examine the AI Reality Gap.
The first case study, augmented Intelligence in Healthcare (Mayo Clinic) is mainly about:

- Diagnostic accuracy
- Clinical decision support
- Operational efficiency

The second case study, cross-model hallucination experiment using:

- OpenAI Chatgpt
- Google AI Mode.

Technical experiments conducted using Python and Google Colab.
UCI Heart Disease dataset used to study data quality degradation.
Random Forest model tested under missing, corrupt, and mixed noise conditions.
Productivity J-Curve mathematically modelled using Stanford AI Index and McKinsey data.
Research compares controlled benchmark performance with real-world deployment outcomes.

Tools and technologies

- Python programming language used for experiments and implementation.
- Google Colab used as the development and execution environment.
- Random Forest Machine Learning model used for data analysis.
- UCI Heart Disease Dataset used for clinical data experiments.
- OpenAI ChatGPT used for reasoning and hallucination testing.
- Google AI Mode used for cross-model evaluation.
- Mathematical modelling used for Productivity J-Curve analysis.
- Research references included Stanford AI Index 2025 and McKinsey AI reports.

Implementation

Two technical experiments were designed to study the AI Reality Gap.
Experiment 1 analyzed how poor-quality data affects AI accuracy.
Preprocessing steps:

- Mean Imputation
- Z-score normalisation $x' = \frac{x-\mu}{\sigma}$
- Stratified Train-Test Split

Noise injection techniques used:

- Missing value noise
- Corruption noise
- Mixed noise

Random Forest model trained and tested at different noise levels.

Experiment 2 implemented the Productivity J-Curve using mathematical modelling.

$$P(t) = B - D \cdot e^{-\delta t} + G \cdot (1^{\rightarrow} e^{-\gamma t})$$

Three adoption scenarios simulated:

- Ideal adoption
- Typical enterprise adoption
- Proof-of-concept trap

All experiments implemented using Python on Google Colab.
Results compared with real-world case studies for practical analysis.

Results

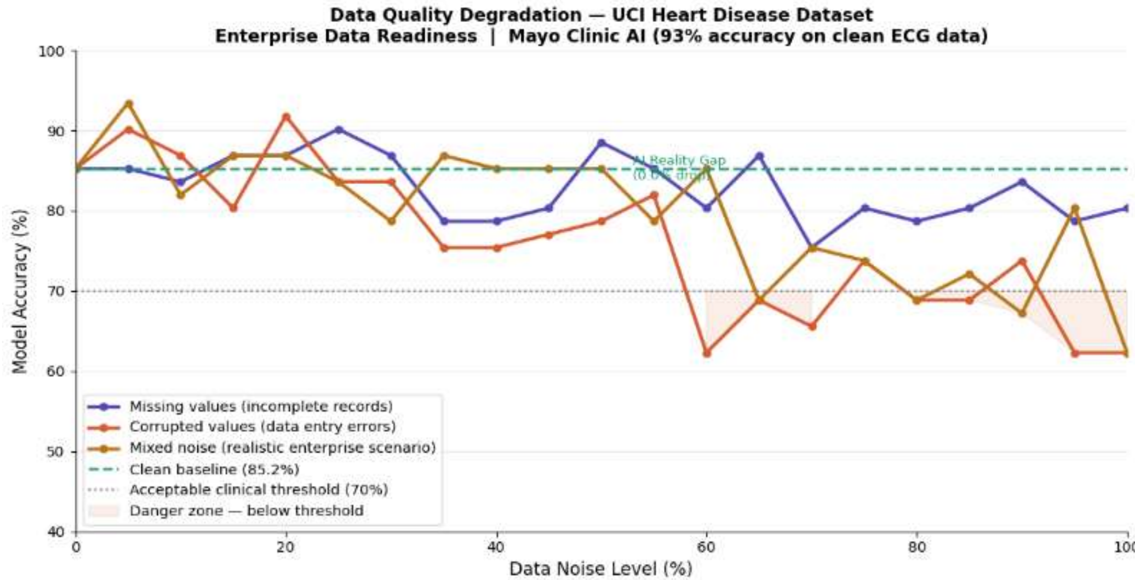
Experiment 1

85.25%
Clean Baseline(Lab Benchmark)

59.02%
Mixed Noise 100% (Real-World)

Data Quality Degradation

Accuracy vs Noise Level- Three Noise Conditions



UCI Heart Disease Dataset
303 patients- 13 features

Key Findings

26.23%

Accuracy drop at full corruption

Results

Experiment 2

Productivity
J-Curve Model

Anchored to Stanford AI Index 2025 & McKinsey AI Survey Data

Productivity Gap

28.1 pts

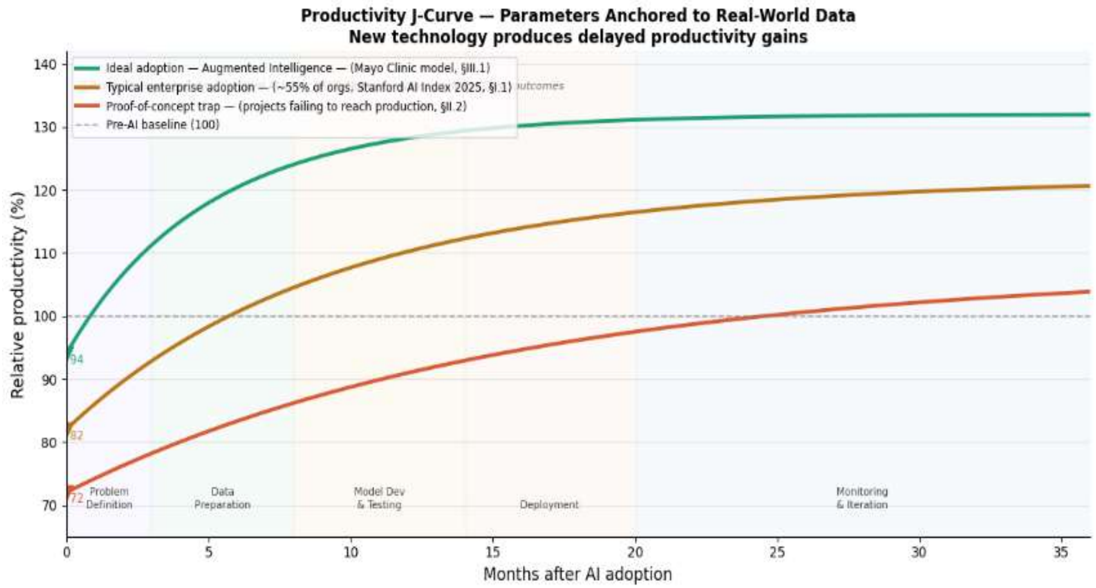
Ideal vs Failed Adoption at Month 36

132.0
Ideal Adoption
(Mayo Clinic Model)

120.6
Typical Enterprise
(55% of Organizations)

103.9
POC Trap
(Failed ROI- 3.9% Gain)

Productivity Over 36 Months- Three Adoption Scenarios



What The Results Tell Us

Synthesis of Experiment 1, Experiment 2 & Case studies

Data Quality

Data Quality Drives the Gap

A robust Random Forest model dropped 26.23% in accuracy — not because the model failed, but because the data around it degraded. The AI Reality Gap is measurable and predictable.

85.25% → 59.02%

Organisational Conditions

Conditions, Not Models, Determine Success

The same technology produced a 28.1-point productivity gap between ideal and failed adoption. Organisational decisions — not model capability — determine the outcome.

132.0 vs 103.9

Hallucination

Hallucination is a Separate Dimension

The cross-model experiment showed AI can fail even with clean inputs. Indirect reasoning triggers hallucination — a failure mode data governance alone cannot fix.

Case Study 2

Conclusion/Future Scope

What This Paper Established

Closing the Gap

Central Conclusion

The AI Reality Gap is a systems problem — not a model problem.

The tools to close it exist. What's missing is recognition that it is the organization's responsibility — not the models'.

Data Quality is Measured

85.25% → 59.02% on real clinical data. Poor data governance costs 26 accuracy points — not model weakness.

Conditions Determine Outcomes

28.1-point productivity gap using the same technology. Organisational decisions, not models, drive success or failure.

Conditions Determine Outcomes

Clean inputs are not enough. Indirect reasoning triggers failure that data governance alone cannot prevent.

Future Scope

RAG vs Vanilla LLM

Measure how much grounding closes the gap

Longitudinal

Track AI adoption across enterprises over 24–36 months

Cross-Domain Noise Testing

Apply degradation methodology to finance, legal & manufacturing

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Thank You